BIRZEIT UNIVERSITY

COMP233 Discrete Mathematics

**Quiz 1 – Chapters 2 and 3 – Section 4**

**100 points | 3 questions | Time: 30 minutes**

Student ID: _____ Student Name: _____

**Answer each part in the space provided below it. Show the requested steps.**

**Question 1 [20 pts]**

Let p = “The battery is charged”, q = “The device turns on”, and r = “The device is faulty.”

a. Formalise each statement in symbols: [10 pts; 5 each]

(i) A charged battery is sufficient for the device to turn on.

(ii) Neither is the battery charged nor does the device turn on.

b. For the proposition “If the battery is charged and the device does not turn on, then the device is faulty”, write its contrapositive and its negation in symbols. In your final answers, ¬ should apply only to single variables. [10 pts; 5 each]

a. (i) Sufficient condition [5 pts]

a. (ii) Neither … nor [5 pts]

b. Contrapositive [5 pts]

b. Negation [5 pts]

**Question 2 [35 pts]**

a. Use a complete truth table to check whether $p \rightarrow (q \vee r)$ and $(p \wedge \neg q) \rightarrow r$ are logically equivalent. Explain your conclusion. [15 pts]

b. Simplify $(p \rightarrow q) \wedge (p \rightarrow \neg q)$ using logical equivalences. Name the law used at each step. Do not use a truth table. [10 pts]

c. From the premises $p \rightarrow q$, $q \rightarrow r$, $\neg r$, and $p \vee s$, derive $s$ using rules of inference. Give the rule and the supporting premises or steps for each derived line. [10 pts]

a. Truth table and conclusion [15 pts]

b. Simplification with named laws [10 pts]

c. Derivation with rules and references [10 pts]

**Question 3 [45 pts]**

a. The domain is all employees of a company. Let M(x) = “x is a manager,” T(x) = “x has completed the security training,” and K(x) = “x holds a key card.” Consider:

“Every manager who has completed the security training holds a key card.”

Write in symbols: (i) the statement, (ii) its contrapositive, (iii) its negation, and also give the negation as an English sentence. ¬ should apply only to predicates. [15 pts; 5 each]

For parts (b)–(d), let S be a set of students and C a set of courses, and let T(x, y) mean “student x has taken course y.”

b. Formalise, using quantifiers with the specified domains: [10 pts; 5 each]

(i) Every course has been taken by at least one student.

(ii) Some student has taken every course.

c. Write the negation of statement (ii) in symbols, with ¬ moved inside so that it applies only to T(x, y). Then state the negation as an English sentence. [10 pts]

d. Let S = {Ali, Dana, Noor} and C = {Logic, Algebra}. The only courses taken are: Ali took Logic, Dana took Algebra, and Noor took Logic. Determine the truth value of statements (i) and (ii) from part (b), and justify each. What does this example show about whether (i) implies (ii)? [10 pts]

a. (i) Statement [5 pts]

a. (ii) Contrapositive [5 pts]

a. (iii) Negation, in symbols and in English [5 pts]

b. (i) [5 pts]

b. (ii) [5 pts]

c. Negation of (ii), in symbols and in English [10 pts]

d. Truth values, justification, and conclusion [10 pts]

**Good luck!**